FACULTY OF COMPUTING
UTM Johor Bahru

SECP3213: BUSINESS INTELLIGENCE

2025/2026 – SEMESTER 2

SECTION 1

PROJECT PHASE 3:

Visualisation and Dashboard

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1.0 SYSTEM ARCHITECTURE & DESIGN FRAMEWORK

To bridge backend data preparation and descriptive business analytics, the processed datasets from Alteryx (olist_logistic_analysis.csv and olist_revenue_analysis.csv) were connected directly to Microsoft Power BI. The visualization layer was architected into a cohesive system across three distinct, user-filtered dashboard viewpoints designed to translate data distributions into commercial action items.

2.0 DASHBOARD PAGE 1: BUSINESS & PRODUCT ANALYSIS OVERVIEW

This dashboard serves as the central hub for macro-level corporate tracking, monitoring high-level KPIs, customer spending brackets, and product category financial performance.

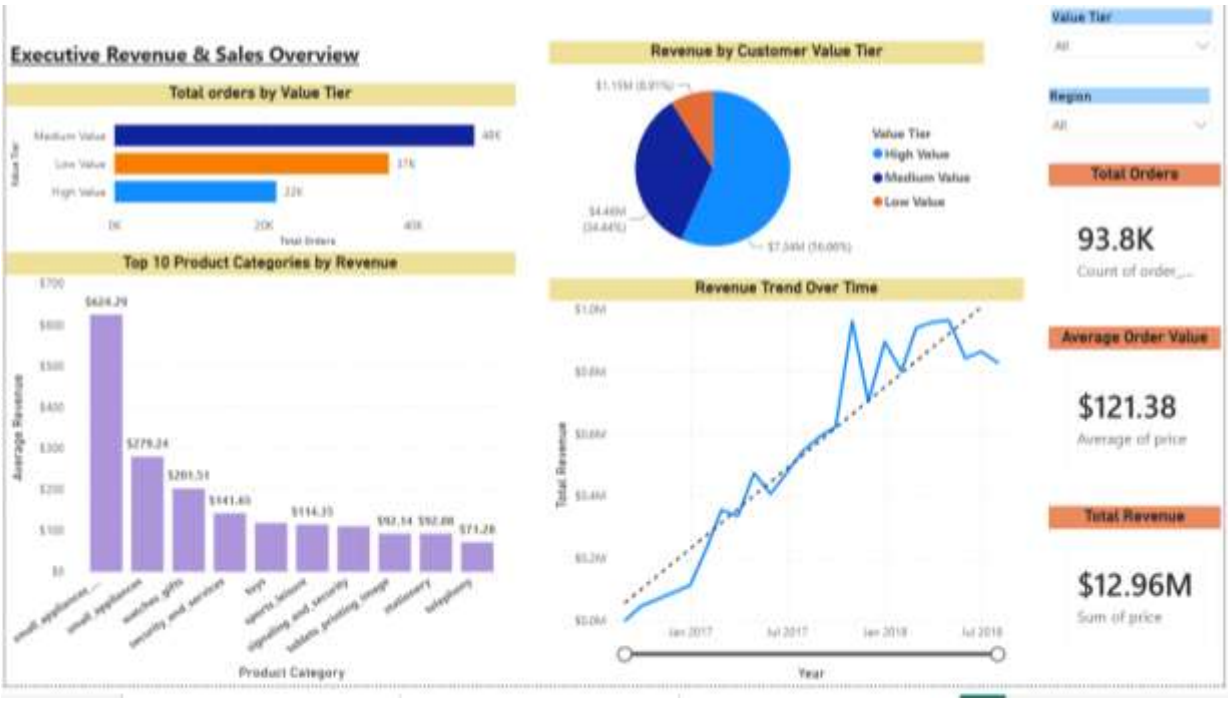

Figure 2.1: Business Overview Dashboard and High-Ticket Product Performance Analysis.

2.1 Charting Justifications & Visualization Best Practices

KPI scorecards were positioned at the top left of the dashboard to align with standard user scanning patterns, thereby providing immediate visibility into core operational figures, namely Gross Revenue ($12.96M), Total Orders (93.8K), and Average Order Value ($121.38). A horizontal bar chart was chosen to display the top ten product categories by revenue, as this format better accommodates the lengthy product category names without needing to tilt the labels. A donut chart was applied to evaluate revenue composition by customer value tier, illustrating that although High-Value accounts represent a minority of raw orders, they account for a disproportionate 56.66% ($7.34M) of total revenue. Finally, interactive slicers and cross-filtering elements were placed at the top right of the dashboard, enabling executives to filter the entire dashboard by calendar year, quarter, and order fulfilment status as required.

2.2 Deep Business Insights & Strategic Recommendations

The findings indicate that Brazil's e-commerce financial performance is driven primarily by high-ticket, durable goods rather than by cheap, high-volume items. Small appliances command an outstanding average sale value of $624.29, followed by small appliances (general) at $279.24 and watches and gifts at $201.51. A significant revenue gap exists between these premium categories and lower-margin categories such as toys and stationery, both of which fall below $115 in average sale value.

Based on these findings, it is recommended that a High-Value Customer Retention Programme be implemented, offering tier-based loyalty perks, faster order handling, and priority service. As this segment generates over half of total corporate revenue from less than a quarter of the order base, even a modest upgrade in the proportion of Medium-Value buyers migrating into this tier would be expected to meaningfully lift total revenue without a corresponding increase in customer acquisition costs. It is further recommended that marketing budgets be redirected to support product bundling and upsell prompts at checkout within the appliances and watches categories, with the objective of raising average basket size beyond the current benchmark of $121.38.

3.0 DASHBOARD PAGE 2: REVENUE STORY & MACRO-TREND ANALYSIS

This section tracks continuous corporate growth vectors, analysing purchase frequencies and income distributions across time series.

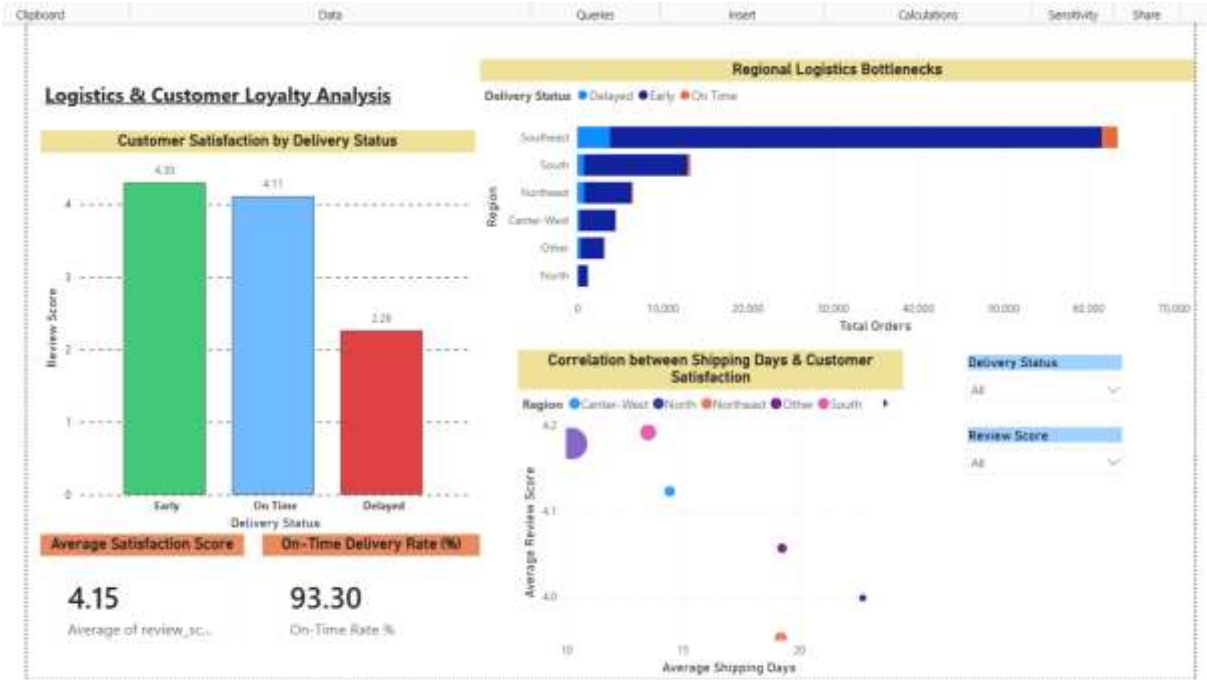

Figure 3.1: Revenue Story Line-Chart Matrix and Spending Distributions.

3.1 Charting Justifications & Visualization Best Practices

The dual-axis trend line chart was selected as the standard visual approach for representing time-series data. This chart maps a continuous monthly timeline along the X-axis, while two Y-axes track sales revenue and total transaction volume concurrently, allowing both metrics to be assessed within a single visual frame. A dashed linear trendline is superimposed on the data to convey the overall trajectory of business performance, enabling the underlying growth pattern to be distinguished from short-term monthly fluctuations.

To examine the distribution of order volumes relative to revenue contribution, a pair of donut charts was presented side by side rather than consolidated into a single visual. The first chart illustrates the composition of order volumes across value tiers, comprising Medium tier orders at 45.2%, Low tier orders at 34.47%, and High tier orders at 20.33%. The second chart presents the corresponding distribution of revenue shares, in which the High tier alone accounts for 56.66% of total revenue. When viewed in conjunction, these two charts reveal a disproportionate relationship between order volume and revenue generation, indicating a structural income inequality across the user base, whereby a relatively small proportion of high-value orders contributes a majority share of total revenue.

3.2 Deep Business Insights & Strategic Recommendations

The data shows a pattern of slow, steady growth, with a clear jump in revenue around November 2017. Following this spike, revenue stabilised at a higher base level rather than reverting to prior levels, eventually reaching a peak of approximately $1.0 million in a single month near January 2018. The sharp decline observed at the tail end of 2018 does not reflect operational decline; rather, it is a reporting artifact resulting from data truncation, as a portion of orders placed during this period remained in transit at the time of extraction and had not yet been recorded as completed.

Based on these findings, it is recommended that the underlying drivers of the November 2017 spike be investigated further to determine whether it resulted from a platform-wide promotional event, a targeted marketing campaign, or organic seasonal demand. If a specific cause can be identified, this strategy could be developed into a structured promotional calendar, which would help reduce month-to-month revenue volatility through planned, repeatable interventions rather than relying on isolated surges. Furthermore, given that only 20.33% of the order base accounts for 56.66% of total revenue, it is recommended that premium product collections and tiered loyalty programmes be introduced to further capitalise on the spending capacity of high-value accounts and to maximise revenue contribution from this customer segment.

4.0 DASHBOARD PAGE 3: LOGISTICS STORY & REGIONAL BOTTLENECKS

This dashboard evaluates supply chain performance, matching physical delivery delay distributions to customer review patterns across Brazil's regions.

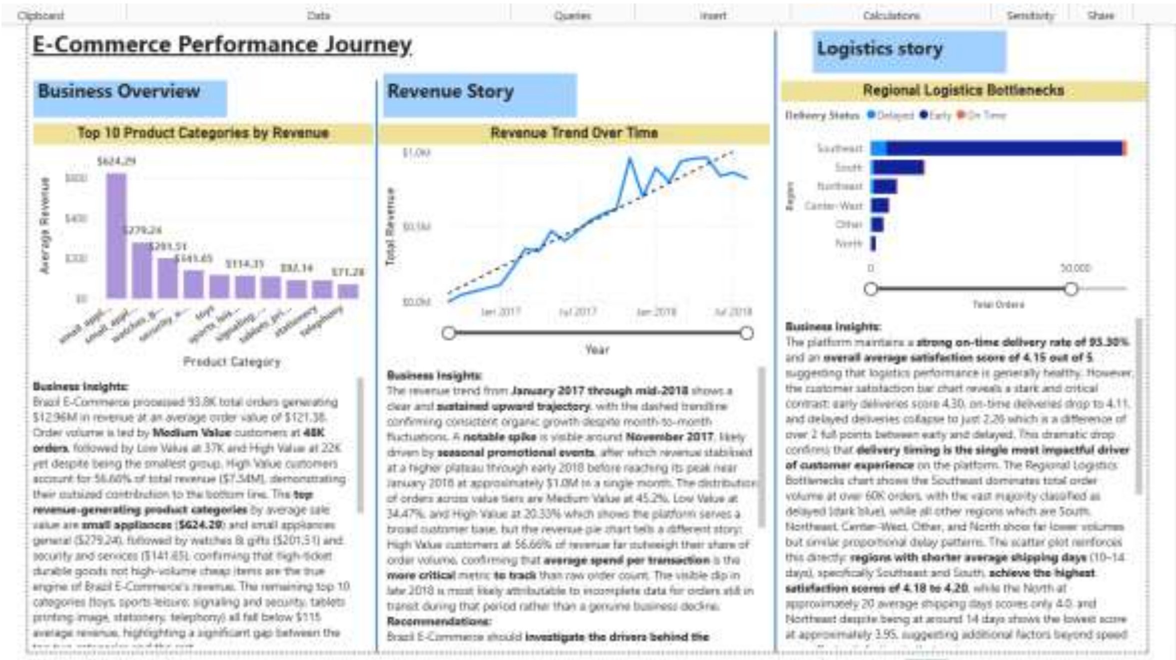

Figure 4.1: Logistics Metrics Dashboard and Regional Shipping Scatter Metrics.

4.1 Charting Justifications & Visualization Best Practices

A clustered stacked column chart was used to represent regional delivery volume alongside delay brackets. This chart groups total order distributions by geographical region, namely Southeast, South, Northeast, Center-West, Other, and North, with each column further subdivided into colour-coded status brackets distinguishing delayed orders from on-time shipments. This structure allows regional order volumes and their associated delivery performance to be assessed simultaneously, making it easier to identify regions that are experiencing a disproportionately high number of delays.

An analytical scatter plot with an overlaid trendline was used to examine the relationship between shipping duration and customer satisfaction. Average shipping duration, measured in days, is plotted along the X-axis, while average review score is plotted along the Y-axis. This visualisation substantiates the moderate negative relationship identified through the Pearson correlation node in Alteryx ($r = -0.3357$), illustrating that as shipping duration increases, average review scores tend to decline correspondingly.

Lastly, a satisfaction status bar chart was constructed using three contrasting colour states to categorise delivery timing classes against average customer ratings. This chart enables a direct comparison of customer satisfaction levels across distinct delivery timing categories, thereby clarifying the extent to which delivery punctuality influences overall customer sentiment.

4.2 Deep Business Insights & Strategic Recommendations

Although the system reflects an overall healthy on-time delivery rate of 93.30% and an average customer satisfaction score of 4.15 out of 5, a breakdown of this score by delivery timing reveals a substantial disparity in customer sentiment. Early shipments achieve a notably high score of 4.30, on-time deliveries register at 4.11, while delayed shipments decline sharply to 2.26. This two-point drop shows that delivery delay is the single biggest factor behind negative customer reviews in the dataset. Regionally, the Southeast accounts for the largest share of total order volume, exceeding 60,000 orders, which correspondingly produces the highest absolute volume of delayed shipments, whereas the North region records the longest average shipping duration, at approximately 20 days.

Given these findings, minimising transit delays should be treated as the top operational priority to protect the company's reputation. It is recommended that secondary regional fulfilment hubs be established and that partnerships be formed with local last-mile carriers within the Southeast region, to mitigate demand pressure in this high-volume area. With respect to the Northeast, where customer ratings remain comparatively low despite moderate shipping windows, it is recommended that a review be conducted to identify non-logistical friction points, such as items damaged in transit or inaccurate product descriptions, as these factors may be contributing to dissatisfaction independent of delivery speed. Finally, a low-cost strategy worth consideration involves extending the delivery dates communicated to customers by one to two days. This adjustment would reframe standard shipments as early arrivals, potentially raising customer satisfaction scores from the 4.11 tier toward the 4.30 tier, without requiring any change to actual warehouse or logistics operations.